SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Geotechnical Engineering: Soil Mechanics & Foundation Engineering

Department: CIVIL	**Course / Subject:** Geotechnical Engineering (CE601PC)
Semester: Semester 6	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: 3-phase soil relationships, Atterberg limits, Permeability (Darcy's Law), Consolidation settlement, Shear strength (Mohr-Coulomb), and Terzaghi bearing capacity.

Unit 1: Physical Properties of Soils

Three-phase diagram, Void ratio, Porosity, Degree of saturation, Water content, Specific gravity, Grain size distribution sieve analysis, Soil classification (USCS/IS).

Unit 2: Soil Permeability & Seepage

Darcy's law, Constant head and falling head permeability tests, Flow net properties, Seepage discharge calculation, Critical hydraulic gradient, Quick sand condition.

Unit 3: Compaction & Consolidation

Standard and Modified Proctor compaction tests, Optimum Moisture Content (OMC), Terzaghi's 1-D consolidation theory, Compression index, Pre-consolidation pressure.

Unit 4: Shear Strength of Soils

Mohr-Coulomb failure criterion, Direct shear test, Unconfined compression test, Triaxial compression test (UU, CU, CD conditions), Pore pressure coefficients.

Unit 5: Bearing Capacity & Shallow Foundations

Terzaghi's bearing capacity theory for strip, circular, square footings, General vs Local shear failure, Water table effect corrections, Standard Penetration Test (SPT).